

Q1)

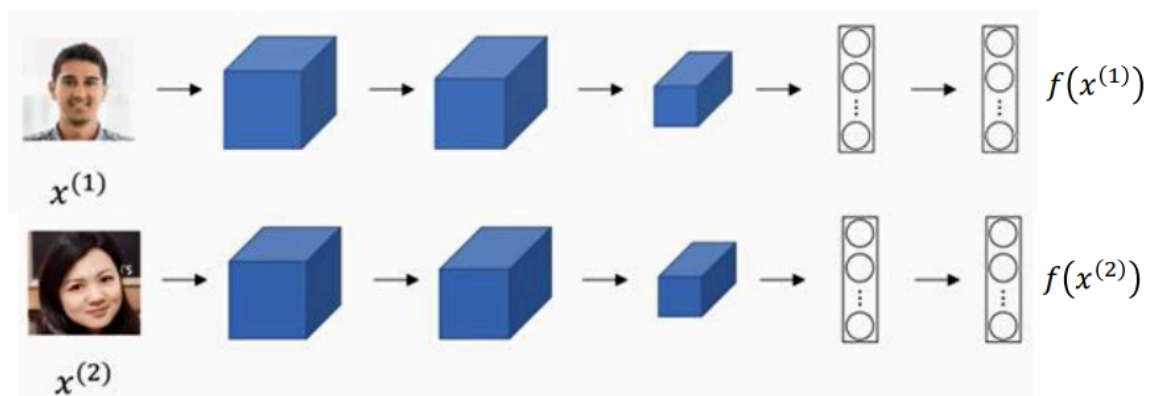
A)

- I. The image at different scales and smoothing used for scale-invariant feature extraction
- II. An approximation of the laplacian used for edge detection with less computational power
- III. Enables rotation normalization later which provides rotation invariance
- IV. Puts all extracted windows in the same scale and orientation

B)

Architecture:

Siamese Network



$$d(x^{(1)}, x^{(2)}) = \|f(x^{(1)}) - f(x^{(2)})\|_2^2$$

If $x^{(i)}, x^{(j)}$ are the same person, $\|f(x^{(i)}) - f(x^{(j)})\|_2^2$ is small.

If $x^{(i)}, x^{(j)}$ are different persons, $\|f(x^{(i)}) - f(x^{(j)})\|_2^2$ is large.

(Lec 8, Slide 48)

Learning Procedure:

- 1) **Input:** Triplets of an anchor image (training image), a positive image (*image from the same class as anchor*) and a negative image (*image from a different class than the anchor*)
- 2) **Learning Objective:** Minimize distance between anchor and positive images and maximize distance between anchor and negative images
- 3) **Loss Function:** Triplet loss

$$L(A, P, N) = \max(\|f(A) - f(P)\|_2^2 - \|f(A) - f(N)\|_2^2 + \alpha, 0)$$

Q2)

1. F
2. T (Lec 4, slide 64)
3. مش علينا
4. F , Semantic segmentation & object detection
5. T
6. T
7. F, each cell can only have 1 class
8. T
9. مش علينا

Q3)

A) مش علينا

B)

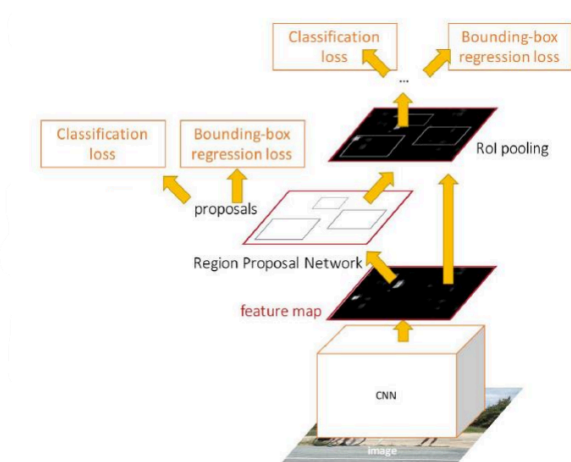
A model that can be used is **Naive Bayes** as follows:

- 1) Extract Features x from images (e.g. *Color histogram*)
- 2) Calculate $p(B | x) = p(x | B) p(B)$ where B represents the *Background*
- 3) Calculate $p(F | x) = p(x | F) p(F)$ where F represents the *Foreground*
- 4) Classify the pixel as B if $p(B | x) > p(F | x)$

Q4)

A)

- I. ROI pooling layer is used to resize/wrap the region proposals to a fixed size



II.

III.

Instead of using Selective search the Faster-RCNN uses a Region Proposal Network to learn RoI's

B)

i.

- Find features (color, gradients, texture, etc.).
- Initialize windows at individual feature points.
- Perform **mean shift algorithm** for each window until convergence.
- Merge windows that end up near the same "peak" or mode.

Mean Shift Algorithm

1. Choose a search window size.
2. Choose the initial location of the search window.
3. Compute the mean location (centroid of the data) in the search window.
4. Center the search window at the mean location computed in Step 3 (shift)
5. Repeat Steps 3 and 4 until convergence.

Parameters:

- Windows size

ii.

- Very large window size is **fast but not accurate** and can lead to mistakenly segmenting of multiple objects as one object
- small window size is **slower but more accurate** but very small size can lead to noise being detected as an object

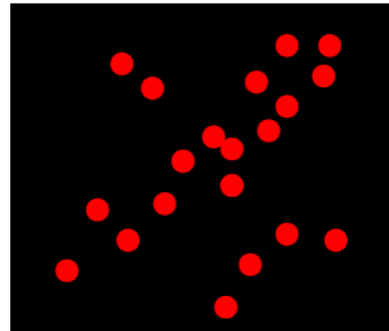
Q5)

A)

I.

RANSAC

1. **Sample** (randomly) the number of points required to fit the model.
2. **Solve** for model parameters using sample.
3. **Score** by the fraction of *inliers* within a preset threshold of the model.
4. **Repeat** 1-3 until the best model is found with high confidence.



(lec 6, slide 24)

II.

$$e = 1 - \text{inliers} = 1 - 0.6 = 0.4$$

$$P = 0.9$$

$$s = 2 \text{ (fitting a line requires min. 2 points)}$$

$$N > \frac{\log(1-p)}{\log(1-(1-e)^s)} = \frac{\log(1-0.9)}{\log(1-(1-0.40)^2)} = 5.15 \approx 6$$

B)

i.

- **Semantic gap:** is the difference between user intent and what could be extracted from the image

(Lec 8, slide 35)

ii.

$$AP_k = \frac{1}{GTP} \sum_{i=1}^K \frac{TP_{seen}}{i}$$

- $k = 6$ (results count)
- $GTP = 3$ (Total positives (*cat images*) in database. given in question)

Model 1:

(cat, dog, dog, cat, cat, car,....)

$$AP_6 = \frac{1}{3} \left(\frac{1}{1} + \frac{0}{2} + \frac{0}{3} + \frac{2}{4} + \frac{3}{5} + \frac{0}{6} \right) = 0.7$$

Model 2:

(cat, cat, cat, dog, car, car,....)

$$AP_6 = \frac{1}{3} \left(\frac{1}{1} + \frac{2}{2} + \frac{3}{3} + \frac{0}{4} + \frac{0}{5} + \frac{0}{6} \right) = 1$$

The **second model** is more accurate.